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Pyrazinamide resistance due to *pncA* gene mutation and its association with treatment outcome among tuberculosis patients of South India- A longitudinal observational study

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A R T I C L E I N F O

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A B S T R A C T

Introduction: Mycobacterium tuberculosis has been extensively studied for mutations leading to drug resistance. Pyrazinamide is a drug acting on the semi-dormant bacteria that is responsible for relapse of tuberculosis. This drug helped reduce the treatment duration of tuberculosis from nine to six months. However, this drug is not being screened for resistance along with Rifampicin and Isoniazid.

Aims and objectives: This study aimed to estimate the proportion of *pncA* gene mutation among tuberculosis patients and its association between treatment outcomes, clinical characteristics, and phenotypic drug resistance.

Method: ology: A total of 154 samples included 73 drug-resistant and 81 drug-susceptible isolates. The isolates were subjected to DNA extraction and amplification using conventional PCR. The PCR product was sequenced by the Sanger sequencing method, and phenotypic drug susceptibility testing was done using the broth dilution method. The association of this gene with the treatment outcome was done by following up with the patients till the end of the regimen.

Results: None of the drug susceptible tuberculosis patients showed significant non-synonymous mutations. Among the drug-resistant TB patients, seven unique significant mutations out of 73 isolates (9.6%) were distributed among Isoniazid-resistant tuberculosis and Multi-Drug Resistant Tuberculosis isolates. No association was found between the mutations and the clinical characteristics of the subjects harboring these isolates.

Conclusion: This study estimated seven unique mutations in drug-resistant tuberculosis and none in drug-sensitive tuberculosis. Isolates harboring was not significantly associated with the participant's treatment outcome and other clinical characteristics. The pyrazinamide resistance testing by the phenotypic and genotypic methods was found to be in concordance.

1. Introduction

Mycobacterium tuberculosis (MTB) has been studied extensively for genetic mutations that confer resistance to drugs used to treat Tuberculosis (TB). The gene *pncA* is one such gene that encodes for pyrazinamidase (PZAse) and helps in the activation of pyrazinamide (PZA) to its active form pyrazinoic acid (POA). Thus, mutation in this gene is linked to PZA resistance.[1]

The drug pyrazinamide and its resistance are being explored in research as this drug mainly acts on slow-growing and persisting bacteria responsible for the relapse of TB. This sterilizing action of pyrazinamide, along with rifampicin, served as the reason for reducing the duration of the treatment regimen from nine to six months.[2]

The drug pyrazinamide is a part of different TB regimens such as Drug susceptible (DS-TB) Tuberculosis regimen, INH-mono/poly drug-resistant TB regimen, and Bedaquiline containing shorter oral Multi-Drug Resistant (MDR) TB regimen.[3] Apart from all bactericidal first-line drugs, this drug is not screened routinely for resistance in the initial phase of the therapy. This is due to a lack of studies reporting resistance to this drug and its burden in patients with TB, and also lacks

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an approved molecular marker for resistance. Thus, the use of this drug without screening for resistance for a usual regular duration remains irrational.

Currently, the WHO recommends Liquid Culture-Drug Susceptibility Test (LC-DST) method to look for resistance in pyrazinamide, which has a drawback of increased incidence of false positive results due to sub-optimal activity of pyrazinamide invitro conditions.[4]

Notably, the proportion of this genotypic resistance to pyrazinamide by the *pncA* mutations was higher among Drug-resistant tuberculosis (DR-TB) isolates, ranging around 40–60%.[5–8] Pyrazinamide resistance and failure to pick the drug resistance in the early part of the treatment regimen play some vital roles in treatment failure.[9] Thus, measures to screen for resistance to pyrazinamide in the early phase of the treatment can significantly help in addressing treatment failure patients among DR TB patients, which was 26% compared to 8.7% among DS-TB patients according to the recent India TB Report 2023.[10]

Though India has a high prevalence of tuberculosis, there is still a lacuna in knowledge regarding mutations in *pncA* and the proportion of mutations among different regimen categories of tuberculosis, such as drug-susceptible and DR-TB, and its association with treatment outcomes.

Hence, in this study, we aimed to address this lacuna by quoting the proportion of *pncA* mutation in drug-susceptible and DR-TB patients. The subjects were also followed up until the end of the regimen to record the treatment outcome associated with the *pncA* mutation profile.

2. Methodology

2.1. Design and setting

This longitudinal observational study was conducted in Pondicherry, Union territory of India. The sample collection was done in the Intermediate Reference Laboratory (IRL), Puducherry, which receives samples from the districts of Pondicherry and Karaikal. Samples of all notified pulmonary tuberculosis patients who have been bacteriologically confirmed with a positive result for *Mycobacterium tuberculosis,* irrespective of their age and gender, were eligible for the study. Samples of extrapulmonary tuberculosis patients and those enrolled in M/XDR-TB patients started on all oral regimens were excluded from this study.

The data from the Electronic Health Record (EHR) was collected hand in hand while processing the samples. Patients were followed up till the completion of the regimen, and data like age, gender, occupational status, and treatment outcome as cured/treatment completed (favorable) and default/failure/death (unfavorable) were entered in the case report form.

3. Study procedure

3.1. Decontamination of samples

The samples were collected from IRL, and the samples were subjected to pre-extraction processing such as decontamination by 4% NALC- NaOH method, and the decontaminated samples were cultured for 14–21 days using BACTEC MGIT 960 system (Becton Dickinson, Sparks, MD, United States) according to the manufacturer's instructions.[11,12]

3.2. DNA extraction, amplification, and gene sequencing

The cultured MGIT samples were transferred to 15 mL centrifuge tubes and centrifuged at 4000 rpm for 5 min. The pellet obtained was used for extraction using the Qiagen DNA Mini Kit. The extracted samples were then subjected to amplification for which 5 μl of DNA product along with 12.5 μl master mix (Sigma Aldrich), 4.5 μl nuclease-free water, and 1.5 μl of each forward and reverse primer of sequence described below were added to get a 25 μl PCR product.

Forward primer: 5′-GGCGTCATGGACCCTATATC-3′

Reverse primer: 5′-CAACAGTTCATCCCGGTTC -3′.

The PCR was run at 94 °C for 5 min for initial denaturation, followed by 35 cycles of denaturation at 94 °C for 1 min, annealing at 56 °C for 1 min, and extension at 72 °C for 1 min, and final extension was run at 72 °C for 10 min. The PCR products were then subjected to post PCR purification using NucleoSpin Gel and PCR Clean-up, Mini kit, Macherey Nagel.

Sequencing was done to this PCR product to analyze the sequence and mutations in the *pncA* gene. The gene sequence was studied for mutations using a sequence from the NCBI portal as a reference sequence using BioEdit software version 7.2.5.

3.3. Phenotypic drug susceptibility testing

Phenotypic DST was done using the Broth microdilution method proposed by Leite et al., for which 96 well plates with U-shaped bottoms were used. The samples were diluted and inoculated with 100 mcg of pyrazinamide as a cut-off to find their phenotypic susceptibility. The plate was incubated at 37 °C for around 21 days to detect the organism's growth.[13]

3.4. Sample size and data analysis

Considering the expected proportion of pyrazinamide resistance as 10% with 5% absolute precision and 95% CI, the sample size is 139. Assuming a 10% attrition, the corrected sample size is estimated to be 154.

The categorical data were expressed as frequency and proportions, and the continuous data, such as Age and BMI, were expressed as mean with standard deviation (SD). The association between *pncA* and clinical characteristics was done by Chi-squared/Fisher's Exact test using IBM SPSS statistics software version 19. A P-value less than 0.05 was considered statistically significant.

3.5. Ethical issues

Ethical clearance was obtained from the Institutional Ethics Committee, JIPMER. The samples obtained were de-identified and anonymized to maintain the confidentiality of the patients.

4. Results

4.1. Baseline characteristics

This study aimed to estimate the proportions of *pncA* mutation in patients on DS-TB and DR-TB regimens and to find its association with treatment outcomes. A total of 154 isolates were taken for study purposes, of which 81 (52.6%) were on the DS-TB regimen, and 73 (47.4%) were under the DR-TB regimen. The distribution of patients based on their baseline characteristics is mentioned below (Table 1).

4.2. Analysis of mutation pattern in the pncA gene

There were 21 (13.6%) isolates with mutations irrespective of drug-resistant patterns, and they were found to be scattered throughout the coding region of *pncA*. There was a total of 11 unique mutations identified across the *pncA* gene, which included synonymous and non-synonymous mutations distributed in both DS-TB and DR-TB with proportions of 8.6 and 19.2%, respectively (Fig. 1). The characteristics of the observed mutations are given in Table 2.

4.3. Association of pncA mutations with clinical characteristics

The association between the *pncA* mutation and clinical characteristics such as type of cases, whether new or previously treated, early

Table 1
Baseline characteristics of TB patients distributed among Drug-Susceptible Tuberculosis (DS-TB) and Drug-Resistant Tuberculosis groups (DR-TB).

Baseline characteristics	DS-TB	DR-TB	Total
	N = 81, n (%)	N = 73, n (%)	N = 154, n (%)
Age (years)[a]	46 (15.6)	47.5 (13.7)	46.7 (14.7)
Gender			
Male	58 (71.6)	56 (76.7)	114 (74)
Female	23 (28.4)	17 (23.3)	40 (26)
BMI (kg/m^2)[a]	18.2 (3.5)	18.2 (3.6)	18.3 (3.5)
BMI category(kg/m^2)			
Underweight (<18.5)	44 (54.3)	44 (60.3)	88 (57.1)
Normal (18.5–22.9)	29 (35.8)	23 (31.5)	52 (33.8)
Overweight & Obese (>23)	8 (9.9)	6 (8.2)	14 (9.1)
Status at TB diagnosis			
New Patients	77 (95.1)	65 (89)	142 (92.2)
Previously treated patients	4 (4.9)	8 (11)	12 (7.8)
Tobacco	17 (20.5)	14 (19.4)	31 (20.1)
Alcohol	14 (17.3)	15 (20.8)	29 (18.8)
HIV	1 (1.2)	1 (1.4)	2 (1.3)
Diabetes Mellitus	26 (32.1)	33 (45.2)	59 (38.3)
Occupation			
Unemployed	17 (21)	17 (23.3)	34 (22.1)
Skilled	4 (4.9)	8 (11)	12 (7.8)
Unskilled	22 (24.7)	13 (17.8)	33 (21.4)
Labored	40 (49.4)	34 (47.9)	75 (48.7)

BMI – Body Mass Index, DS-TB – Drug-Susceptible Tuberculosis, DR-TB – Drug-Resistant Tuberculosis, TB-Tuberculosis, HIV- Human Immunodeficiency Virus.
[a] Expressed as mean (SD).

sputum conversion status, and treatment outcome are given in Table 4, and they showed no significant association.

4.4. Treatment outcome

The three mutations seen in unfavorable treatment outcomes were 80 (T > C), 395 (G > C), and 392 G ins mutations (see Table 3). The mutation at 80 (T > C) was associated with treatment failure.

Type of cases.

Among the 73 subjects, eight had a previous treatment history, and 65 were newly treated. The proportion of mutated isolate was higher among the previously treated patients (12.5%) than the new cases (9.2%).

4.5. Phenotypic resistance of pyrazinamide

Phenotypic resistance to pyrazinamide by broth dilution method revealed the pyrazinamide resistance in 8 samples at 100 μg/mL concentration of pyrazinamide, of which six (75%) had mutations in the *pncA* gene and two in isolates with no *pncA* mutation as given in Table 5.

5. Discussion

This study included 154 isolates with drug-susceptible and drug-resistant strains and was analyzed for the mutations in the *pncA* gene. The occurrences of these mutations and their association with the treatment outcome were done by following up with the patients till the end of their regimen. None of the DS-TB patients showed significant non-synonymous mutations. Among the drug-resistant TB patients, seven unique mutations were distributed among the Hr-TB and MDR-TB patients. No association was found between the mutations and the clinical characteristics of the subjects harbouring these isolates.

In this study, considering only the non-synonymous mutations as significant mutations, only seven out of 21 were observed, distributed among the 73 patients with DR-TB regimens (9.6%). This DR-TB group includes Hr-TB isolates and MDR-TB isolates. In a study conducted by Mvelase et al. in South Africa, 80 rifampicin-resistant isolates were analyzed for *pncA* mutation, which revealed the proportion of mutation to be 7.5%, similar to our study.[14]

Few studies were conducted in countries other than India to identify *pncA* proportions on DR-TB or MDR-TB isolates, which showed proportions around 40–60%, with most studies showing proportions around 50%.[5,7,8,15–17] One study by Li et al. from Chongqing, China, in 465 drug-resistant isolates showed that 74.4% of isolates had 124 different types of mutations throughout the *pncA* gene.[6]

This considerable variation in proportions of *pncA* gene mutations observed in other studies compared to ours can be attributed to the diluted numbers of MDR-TB strains and the small number of mutations captured among them, which was less than expected. There was also a lack of knowledge regarding the proportion of *pncA* among the Indian

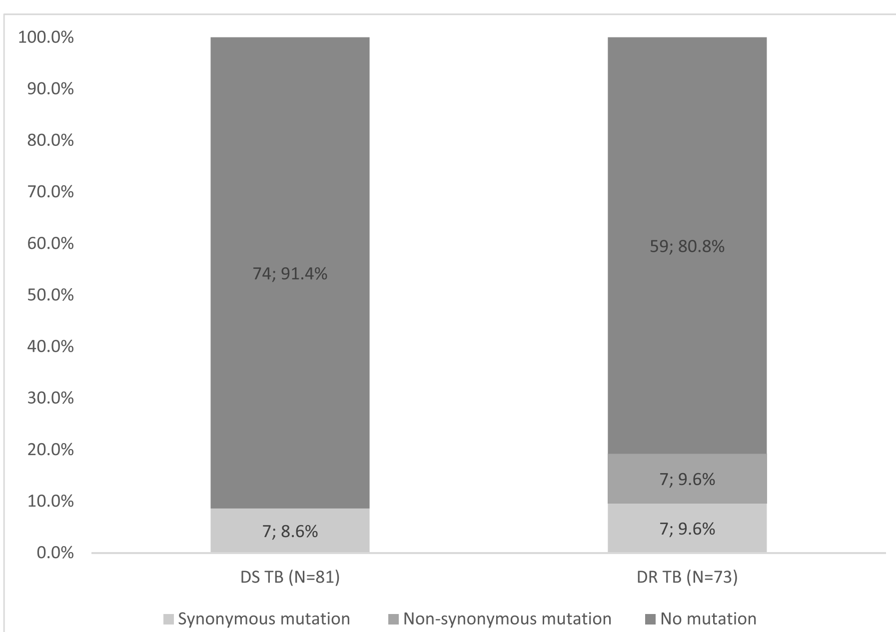

Fig. 1. Proportions of type of mutations among Drug Susceptible tuberculosis (DS-TB) and Drug-Resistant tuberculosis (DR-TB)
The values in the graph are proportions expressed as n; %.

Table 2
Characterization of *pncA* mutations in isolates and their phenotypic drug susceptibility status.

Location of mutation	Nucleotide change	Amino acid change	No of isolates (n = 21)	Phenotypic drug-susceptibility test for pyrazinamide	Drug resistance pattern
Non-synonymous mutations					
80	T > C	Leu27Pro	1	Sensitive	MDR-TB
188	A > G	Asp63Gly	1	Resistant	MDR-TB
226	A > C	Thr76Pro	1	Resistant	Hr-TB
391_392	ins G	Frameshift	1	Resistant	MDR-TB
395	G > C	Gly132Ala	1	Resistant	Hr-TB
395	G > A	Gly132Asp	1	Resistant	Hr-TB
512	C > T	Ala172Val	1	Resistant	MDR-TB
Synonymous mutations					
195	C > T	Synonymous	11	Sensitive	DS-TB, MDR-TB, and Hr-TB
240	C > T	Synonymous	1	Sensitive	Hr-TB
386_395	del ATGTGGTCG	Synonymous	1	Sensitive	DS-TB
420	C > T	Synonymous	1	Sensitive	DS-TB

DS-TB- Drug Susceptible Tuberculosis, MDR-TB- Multi Drug Resistant Tuberculosis, Hr-TB- Isoniazid resistant Tuberculosis.

Table 3
Distribution of *pncA* mutation among different categories of the DR-TB group.

Mutation status		Hr-TB N = 37	MDR-TB N = 36	P-value[a]
pncA mutations	Present **(n = 7)**	2 (5.4)	5 (13.9)	0.261
	Absent **(n = 66)**	35 (94.6)	31 (86.1)	

Hr-TB-Isoniazid resistant TB; MDR TB- Multidrug-resistant TB.
All values are proportions and are expressed as n (%).
[a] Fisher's Exact test.

Table 4
Association of *pncA* mutation with clinical characteristics in drug-resistant tuberculosis patients.

Parameters	*pncA* mutation present n (%)	*pncA* mutation absent n (%)	P-value[a]
Treatment outcome			
Unfavorable (N = 18)	3 (16.7)	15 (83.3)	0.353
Favorable (N = 55)	4 (7.3)	51 (92.7)	
Early sputum conversion status			
Yes (N = 61)	5 (8.2)	56 (91.8)	0.323
No (N = 12)	2 (16.7)	10 (83.3)	
Type of case			
Previously treated patients (N = 8)	1 (12.5)	7 (87.5)	0.573
New Patients (N = 65)	6 (9.2)	59 (90.8)	

All values are proportions expressed as n (%).
[a] Fisher's Exact test.

Table 5
Phenotypic and genotypic mutation status.

Pyrazinamide susceptibility pattern		Phenotypically Resistant (n = 8)	Phenotypically Sensitive (n = 65)	P-value[a]
pncA mutation	Present (n = 7)	6 (75)	1 (1.5)	<0.001
	Absent (n = 66)	2 (25)	64 (98.5)	

All values are proportions, expressed as n (%).
[a] Fisher,s Exact test.

population, which could have been a valid comparator for our study. Another factor is the exclusion of all oral M/XDR-TB regimen that requires 18–21 months of therapy, which was beyond the duration of this study.

Isoniazid-resistance MTB isolates showed a mutation in 5.4% (2 out of 37 isolates) in our study, similar to a study conducted in Hong Kong by Kevin et al. where it was 4.5% (7 out of 157 isolates).[14] Another study in Vietnam showed the proportions as 14.9%, which was relatively higher than ours.[18]

This increased proportion of mutations in previously treated patients is worth considering, as the mutation in *pncA* per se would have led to the recurrence of TB in this group of patients. The above hypothesis is substantiated by the fact that pyrazinamide is active in killing the semi-dormant and persisting bacteria responsible for tuberculosis recurrence.[19] Hence, in patients with relapse of tuberculosis, screening for pyrazinamide resistance might help to achieve a favorable outcome of tuberculosis and prevent further episodes.

The proportion of *pncA* mutations among unfavorable treatment outcomes was 16.7% compared to 7.3% among the favorable treatment group. A study conducted in Tanzania, USA, showed that among the patients who had failed in treatment, 83% had pyrazinamide resistance compared to only 41.6% in patients with successful treatment.[20] A study conducted in South India on treatment failure patients concluded that 78% (39 out of 50) of patients showed a mutation in *pncA*.[21]

Though hypothesizing that mutation affects the treatment outcome, we could find four mutations in patients with favorable treatment outcomes in this study. This discrepancy of having a favorable treatment outcome despite having *pncA* mutation can be due to the effect of other drugs in the regimen.

Further, to discuss the individual significant mutations, they mainly were point substitution mutations (six). The mutation observed in position 226 (A > C) led to the change of amino acid (Threonine to Proline), which was observed in a Pakistan study.[22] Another study from China also reported that the same mutation also showed reduced activity of the pyrazinamidase.[5]

Similarly, there was a substitution mutation in position 188 (A > G) and 395 (G > A), leading to a change in amino acid Aspartate to Glycine and Glycine to Aspartate, respectively. These mutations were reported in a study in China by Li et al., which showed the presence of these mutations.[6]

The other mutations, 80(T > C), 395(G > C), 512(C > T), and 392 (ins G), were reported in the WHO catalog of 2021, where they have listed mutations in *Mycobacterium tuberculosis* complex and their association with drug-resistance.[23]

Among the 73 drug-resistant isolates, 70 showed concordant results with genotypic and phenotypic susceptibility testing. Two isolates did not have any mutation but showed phenotypic resistance. Similarly, in other studies, isolates showed resistance to phenotypic DST but lacked the mutation in the *pncA* gene.[5] This can be due to false positive results, which can be due to factors such as acidic pH and inoculum size influencing the action of pyrazinamide.[9,24,25]

The main strength of this study lies in the study design, where characterizing the mutation goes on one side, following up with the patients for the outcome of their treatment, and associating the mutations with treatment outcomes, which helps us to find the clinical burden of this mutation in treating the patients.

This study's main limitation was the fewer mutations observed due to a smaller number of MDR-TB strains. As only DR-TB subjects were considered for analysis, numbers were not powered enough to attain a conclusion. To overcome these challenges, studies with larger sample sizes focussing mainly on M/XDR-TB subjects or treatment failure subjects may be considered.

Another limitation was detecting pyrazinamide phenotypic susceptibility using the broth dilution method, where WHO recommends LC-DST for phenotypic susceptibility testing of drugs. However, our study did not consider this due to the limited resources and time.

By focussing on mutational studies among the treatment failure group of subjects, we can impact tuberculosis management. With considerable research data on specific mutations, this gene can be used as a marker to predict the diagnosis of resistance among patients and to coin out the most appropriate regimen for pyrazinamide-resistant subjects.

This study identified the proportions of *pncA* mutation in DR-TB as 9.6%, and no significant mutation was seen among DS-TB isolates. The mutations in the *pncA* gene were not associated with the clinical characteristics. The pyrazinamide resistance testing by the phenotypic and genotypic methods was found to be in concordance. More future studies with larger sample sizes and focussing mainly on drug-resistant and treatment failure isolates might help in including *pncA* mutation as a marker for early detection of pyrazinamide resistance.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

1. Via LE, Savic R, Weiner DM, et al. Host-mediated bioactivation of pyrazinamide: implications for efficacy, resistance, and therapeutic alternatives. *ACS Infect Dis [Internet]*. 2015;1:203–214. Available from: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4467917/. Accessed June 29, 2023.
2. Gopal P, Grüber G, Dartois V, Dick T. Pharmacological and molecular mechanisms behind the sterilizing activity of pyrazinamide. *Trends Pharmacol Sci*. 2019;40:930–940.
3. Guidelines for PMDT in India.pdf [Internet]. [cited 2023 May 18]. Available from: https://tbcindia.gov.in/WriteReadData/l892s/8368587497Guidelines%20for%20PMDT%20in%20India.pdf.
4. WHO consolidated guidelines on tuberculosis: module 3: diagnosis: rapid diagnostics for tuberculosis detection, 2021 update. Available from: https://www.who.int/publications-detail-redirect/9789240029415. Accessed June 21, 2023.
5. Shi D, Zhou Q, Xu S, Zhu Y, Li H, Xu Y. Pyrazinamide resistance and pncA mutation profiles in multi-drug resistant Mycobacterium. *Tuberculosis. Infect Drug Resist*. 2022;15:4985–4994.
6. Li K, Yang Z, Gu J, Luo M, Deng J, Chen Y. Characterization of pncA mutations and prediction of PZA resistance in Mycobacterium tuberculosis clinical isolates from chongqing, China. *Front Microbiol [Internet]*. 2021;11, 594171. Available from: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7832174/. Accessed July 5, 2023.
7. Shrestha D, Maharjan B, Thapa J, et al. Detection of mutations in pncA in Mycobacterium tuberculosis clinical isolates from Nepal in association with pyrazinamide resistance. *Curr Issues Mol Biol [Internet]*. 2022;44:4132–4141. Available from: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9497661/. Accessed July 3, 2023.
8. Ei PW, Mon AS, Htwe MM, et al. Pyrazinamide resistance and pncA mutations in drug resistant Mycobacterium tuberculosis clinical isolates from Myanmar. *Tuberc Edinb Scotl*. 2020;125, 102013.
9. Siddiqi S, Ahmed A, Asif S, et al. Direct drug susceptibility testing of Mycobacterium tuberculosis for rapid detection of multi-drug resistance using the bactec MGIT 960 system: a multicenter study. *J Clin Microbiol [Internet]*. 2012;50:435–440. Available from: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3264138/. Accessed July 3, 2023.
10. TB Annual report-2023_23- 03-2023_LRP.pdf. Available from: https://tbcindia.gov.in/WriteReadData/l892s/5646719104TB%20AR-2023_23-%2003-2023_LRP.pdf. Accessed July 7, 2023.
11. Bradner L, Robbe-Austerman S, Beitz DC, Stabel JR. Chemical decontamination with N-Acetyl-l-Cysteine–Sodium hydroxide improves recovery of viable Mycobacterium avium subsp. paratuberculosis organisms from cultured milk. *J Clin Microbiol [Internet]*. 2013;51:2139–2146. Available from: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3697694/. Accessed October 8, 2023.
12. Krüüner A, Yates MD, Drobniewski FA. Evaluation of MGIT 960-based antimicrobial testing and determination of critical concentrations of first- and second-line antimicrobial drugs with drug-resistant clinical strains of Mycobacterium tuberculosis. *J Clin Microbiol*. 2006;44:811–818.
13. Leite CQ, Beretta AL, Anno IS, Telles MA. Standardization of broth microdilution method for Mycobacterium tuberculosis. *Mem Inst Oswaldo Cruz*. 2000;95:127–129.
14. Ng KKM, Yip PCW, Leung PKL. Prevalences of levofloxacin resistance and pncA mutation in isoniazid-resistant Mycobacterium tuberculosis in Hong Kong. *Hong Kong Med J Xianggang Yi Xue Za Zhi*. 2021;27:421–427.
15. Che Y, Bo D, Lin X, Chen T, He T, Lin Y. Phenotypic and molecular characterization of pyrazinamide resistance among multidrug-resistant Mycobacterium tuberculosis isolates in Ningbo, China. *BMC Infect Dis [Internet]*. 2021;21:605. Available from: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8228925/. Accessed July 1, 2023.
16. Liu Q, Yang D, Qiu B, et al. Drug resistance gene mutations and treatment outcomes in MDR-TB: a prospective study in Eastern China [Internet] *PLoS Negl Trop Dis*. 2021;15, e0009068. Available from: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7850501/. Accessed July 3, 2023.
17. Sodja E, Koren S, Toplak N, Truden S, Žolnir-Dovč M. Next-generation sequencing to characterise pyrazinamide resistance in Mycobacterium tuberculosis isolates from two Balkan countries. *J Glob Antimicrob Resist [Internet]*. 2022;29:507–512. Available from: https://www.sciencedirect.com/science/article/pii/S221371652100254X. Accessed July 5, 2023.
18. Huy NQ, Lucie C, Hoa TTT, et al. Molecular analysis of pyrazinamide resistance in Mycobacterium tuberculosis in Vietnam highlights the high rate of pyrazinamide resistance-associated mutations in clinical isolates. *Emerg Microbes Infect*. 2017;6:e86.
19. Zhang Y, Scorpio A, Nikaido H, Sun Z. Role of acid pH and deficient efflux of pyrazinoic acid in unique susceptibility of Mycobacterium tuberculosis to pyrazinamide. *J Bacteriol*. 1999;181:2044–2049.
20. Juma SP, Maro A, Pholwat S, et al. Underestimated pyrazinamide resistance may compromise outcomes of pyrazinamide containing regimens for treatment of drug susceptible and multi-drug-resistant tuberculosis in Tanzania. *BMC Infect Dis*. 2019;19:129.
21. Muthaiah M, Jagadeesan S, Ayalusamy N, et al. Molecular epidemiological study of pyrazinamide-resistance in clinical isolates of Mycobacterium tuberculosis from South India. *Int J Mol Sci [Internet]*. 2010;11:2670–2680. Available from: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2920559/. Accessed July 3, 2023.
22. khan MT, Malik SI, Ali S, et al. Pyrazinamide resistance and mutations in pncA among isolates of Mycobacterium tuberculosis from Khyber Pakhtunkhwa, Pakistan. *BMC Infect Dis [Internet]*. 2019;19:116. Available from: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6364397/. Accessed July 1, 2023.
23. Catalogue of mutations in Mycobacterium tuberculosis complex and their association with drug resistance. Available from: https://www.who.int/publications-detail-redirect/9789240028173. Accessed July 6, 2023.
24. Chedore P, Bertucci L, Wolfe J, Sharma M, Jamieson F. Potential for erroneous results indicating resistance when using the Bactec MGIT 960 system for testing susceptibility of Mycobacterium tuberculosis to pyrazinamide. *J Clin Microbiol*. 2010;48:300–301.
25. Piersimoni C, Mustazzolu A, Giannoni F, Bornigia S, Gherardi G, Fattorini L. Prevention of false resistance results obtained in testing the susceptibility of Mycobacterium tuberculosis to pyrazinamide with the bactec MGIT 960 system using a reduced inoculum. *J Clin Microbiol [Internet]*. 2013;51:291–294. Available from: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3536208/. Accessed July 3, 2023.